

MD. TAWHID MOSTAFA

Software Engineer | AI & Computer Vision Engineer

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Professional Summary

Software Engineer specializing in **applied AI & computer vision**, with production experience across the full stack. Currently developing **AI Tutors** and leading **PateShaba** at NZTECH. Designed, trained, and benchmarked **3D CNN, ConvLSTM, and LSTM** models on a 613-video physiotherapy dataset under a Ministry of Science & Technology research grant, co-authoring a peer-reviewed publication in *Data in Brief* (Elsevier). Comfortable owning a feature end-to-end — from React/Next.js frontend and Django/ASP.NET Core backend, through to training, evaluating, and deploying computer-vision models with TensorFlow, OpenCV, and YOLOv8.

• Published AI Researcher

Data in Brief, Elsevier (2026)

• Full-Stack + Applied AI

React/Django to TensorFlow/YOLO

• Production Ownership

Head Developer, PateShaba

Technical Skills

AI / Machine Learning / Computer Vision: TensorFlow, Keras, OpenCV, CNN, 3D CNN, ConvLSTM, LSTM, YOLOv8, YOLOv11, Model Training & Evaluation, RMSE/MAD Metrics

Web & Backend: React, Next.js, Django, Django REST Framework, ASP.NET Core MVC, REST API Design, JWT Authentication

Languages: Python, JavaScript, Java, C#, C++

Databases: MySQL, PostgreSQL, SQLite, Supabase

Tools & Version Control: Docker, Docker Compose, Git, GitHub, Jupyter Notebook, Visual Studio

Languages Spoken: Bangla (Native), English (Professional Working Proficiency)

Professional Experience

Software Engineer

NZTECH, Dhaka, Bangladesh

January 2026 – Present

- Develop and ship user-facing and admin features for **AI Tutors**, a SaaS platform built on **React** and **Supabase**, working directly from product requirements to production release.
- Serve as **Head Developer** for **PateShaba**, responsible for architecture decisions, module implementation, debugging, and end-to-end delivery.
- Design frontend workflows and data-integration logic between the application and backend services, prioritizing reliability across iterative releases.

Projects

AI-Monitored Physiotherapy Home Assistant

[GitHub](#)

- Engineered a full patient / doctor / admin platform (**Django, TensorFlow/Keras, Bootstrap**) that serves the trained physiotherapy assessment model in production, including automated report generation.
- Deployed the best-performing **3D CNN** model (selected after benchmarking against ConvLSTM and LSTM) to score uploaded exercise videos in real time.

AI Vision Image Analysis Platform

[GitHub](#)

- Built a full-stack object-detection application (**Next.js** frontend, **Django REST Framework** backend) with **JWT authentication** in HTTP-only cookies and a MySQL 8.0 backend.
- Integrated **YOLOv8/YOLOv11** for real-time object detection, returning annotated images with a sortable results table (class, confidence, bounding box); containerized the full stack with **Docker Compose** for reproducible deployment.
- Added a **Gemini 2.5 Flash**-powered conversational assistant so users can query detection results in natural language.

- Designed a 3-role (admin / teacher / student) school management system for Classes 4–8 using **Django** and **MySQL**, covering assignments, grading, attendance, scheduling, and notices.

Research Experience & Publication

AI-Monitored At-Home Physiotherapy Assessment System

June 2022 – September 2025

Special Research Grant (SRG-222431) – Ministry of Science & Technology, Bangladesh

Full-time research role, undertaken immediately after completing undergraduate coursework

- Trained and benchmarked **3D CNN, ConvLSTM, and LSTM** architectures on a **613-video** labeled dataset (adduction, back extension, hip flexion) to output continuous exercise-quality scores from 2D video.
- Evaluated models using **RMSE and MAD across 9 physiotherapy exercises**; identified 3D CNN as the most accurate and stable architecture.
- Led dataset preparation — frame sampling, resizing, normalization — as part of a 4-member government-funded research team.
- Co-authored **MobiPhysio: A 2D Video Dataset of Physiotherapy Exercises for AI-Driven Assessment and Monitoring**, *Data in Brief*, Elsevier, 2026. DOI: [10.1016/j.dib.2026.112635](https://doi.org/10.1016/j.dib.2026.112635)

Education

B.Sc. in Computer Science & Engineering — Stamford University Bangladesh

2018–2022

CGPA: **3.70/4.00** (Result published September 2024, after completing the full-time research role listed above)

HSC, Science — Masjid Mission Academy, Rajshahi

2017

GPA: 3.58/5.00

SSC, Science — Jhikra High School, Rajshahi

2015

GPA: 4.44/5.00

Certifications & Training

- ASP.NET Core MVC Professional Training (Ongoing) — BASIS Institute of Technology & Management (BITM)
- Foundational C# — Microsoft, Mar 2025
- Artificial Intelligence Training — EDGE Project (BCC & IIT, Dhaka University), Dec 2023
- 21st Century Employability Skilling Program – Advanced, EDGE Project, Dec 2023
- Programming for Everybody (Python), Coursera — Jan 2021